

Abdulkaleem Khan

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Design Verification Methodology Engineer | Intel | Semiconductor Industry Expert | EDA Tools Expert

Career Objective:

Design Verification Methodology Engineer with over 5 years of experience at Intel, specializing in EDA tool integration, DV automation, and simulation flow optimization. Expert in Synopsys tools including VCS, Verdi, VMS, and VC Execution Manager, with a proven track record in building scalable DV solutions and accelerating chip development cycles. Recognized as a top performer, consistently exceeding expectations while contributing to the tape-out of Ethernet, AI, and IPU chips. Passionate about solving EDA challenges and driving verification productivity through innovation.

EDUCATION:

MTech in Microelectronics & VLSI Design

July 2018-June 2020

National Institute of Technology (NIT), Silchar, India

Project: Developed and deployed an accelerated Gate-Level Simulation methodology that significantly reduced DV setup and run time.

CPI-8.33

B.E in Electronics & Communication Engineering

July 2010 – June 2014

Babaria Institute of Technology, Vadodara

CGPA-8.4

PROFESSIONAL EXPERIENCE:

Design Verification Methodology Engineer (Intel Technology | Bengaluru | Since July 2020).

- Engineered Intel's first Specman-based DV flow, accelerating Ethernet IP verification and contributing to 3+ successful tape-outs.
- Spearheaded DV methodology support for Ethernet and IPU products, enabling over 5 successful tape-ins across multiple silicon platforms.
- Contributed to development and bring-up of Intel's first AI connectivity chip; enhanced DV environment for FNIC architecture.
- Led tool performance optimization, reducing simulation elaboration time by 29%, memory usage by 22%, and test runtime by 31%.
- Specialized in Synopsys toolchain setup and scaling: **VCS, Verdi, VMS, VC Execution Manager**, ensuring seamless integration and license availability across global DV teams.
- Optimized coverage planning and closure using VManager and Verdi; improved verification cycle efficiency by more than 20%.
- Recognized as top performer consistently; exceeded role expectations in every performance cycle and delivered high-impact tooling improvements ahead of schedule.

Design Automation Engineer intern (Intel Technology | Bengaluru | From July 2019 to July 2020).

- Developed and maintained DV simulation and debug automation flows; resolved 50+ tool issues across DV domain.
- Designed and deployed a scalable GLS methodology that cut setup time by more than 30% and improved DV team efficiency across multiple IP teams.

Project Engineer (Physical Research Laboratory (ISRO), Ahmedabad | July 2014 – July 2016).

- Designed and validated Front-End Electronics and peak detection circuits for payloads on Aditya L1 and Chandrayaan-2/3, supporting launch readiness and signal integrity.
- Engineered FPGA-based data acquisition modules for ASPEX payload (SWIS & TEPS) using LabVIEW and Verilog, enabling real-time solar particle data analysis.
- Developed analog front-end systems and calibration circuits for APIXS (rover) and XSM (orbiter) spectrometers, improving energy resolution and spectral accuracy.

Technical Skills:

- **HDLs:** Verilog.
- **HVLs:** SystemVerilog.
- **Methodologies:** UVM, OVM.
- **EDA Tools:** VCS, Verdi, VMS, Xcelium, Questasim, Specman, Indago, VManager.
- **Simulation & Debug:** RTL simulation, GLS, coverage planning & analysis.
- **Scripting Languages:** Python, Perl, Shell.
- **Domains:** DV automation, performance tuning, tool integration, coverage closure.

Key Projects:

- **GLS Optimization (M.Tech Final Year Project):**
Developed a novel GLS methodology that reduced simulation time and improved efficiency for design teams.
- **Microcontroller & GPS-Based Automated Train Speed Control (B.E):**
Built a train automation system to regulate speed using GPS, enhancing safety against overspeeding.
- **ASPEX (Aditya L1 Mission, ISRO):**
Developed analog and digital electronics for the Solar Wind Spectrometer payloads, contributing to India's solar mission.
- **APIXS (rover) and XSM (orbiter) (Chandrayaan-2&3, ISRO):**
Designed Front-End Electronics and peak detector systems for energy spectrometer.

Workshops & Conferences:

- National Space Science Symposium 2016, VSSC – Paper on payload development for future ISRO missions.
- IEEE EDS Mini Colloquium – Microelectronics & VLSI Design.
- INUP Workshop on Nanofabrication Technology – NIT Silchar.
- PLANEX 15th Workshop – Remote Sensing & Analog Studies on Mars and Moon.
- MOM Data Analysis Workshop – Mars Orbiter Mission, PRL Ahmedabad.

Hobbies & Interests:

- Cricket, Table Tennis, Spending time with family.